An Annotated R Script Illustrating Pharmacometric Population Simulation as a Causal Inference Procedure

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0.1 Credit

The following illustration is loosely based on remarks made in Pearl, Glymour and Jewell, *Causal inference in statistics: a primer*. Wiley, 2016. An anonymized version of the data below is analyzed in that manuscript.

0.2 Kidney Stone Treatment Data

We begin by "reverse engineering" the data in Table II of Charig et al., Comparison of treatment of renal calculi by open surgery, percutanesous nephrolithotomy, and extracorporeal shockwave lithotripsy. *BMJ* 1986;**292**:879–882. We restrict attention to the the rows of that table comparing open surgery to percutaneous nephrolithotomy.

```
## Enter tabulated BMJ data
n <- matrix(c(81, 234, 192, 55), nrow = 2)
N <- matrix(c(87, 270, 263, 80), nrow = 2)
surgery <- matrix(rep(c("open", "percutaneous"), 2), nr = 2)
stone_size <- matrix(rep(c("small", "large"), each = 2), nr = 2)

## Reverse engineer primitive BMJ data
dat <- data.frame(
                  Surgery = rep(surgery, N),
                  Size = rep(stone_size, N),
                  Success = do.call("c",
                                    lapply(seq(n),
                                           function(i) c(rep(0, N[i]-n[i]),
                                                         rep(1, n[i]))
                                    )
                  )
)
```

The following tables reproduce information in selected rows of the original tabulation:

```
dat %>% filter(Success == 1) %>% with(table(Surgery, Size))
```

```
##                Size
## Surgery         large small
##   open            192    81
##   percutaneous     55   234
```

```
dat %>% with(table(Surgery, Size))
```

```
##                Size
## Surgery         large small
##   open            263    87
##   percutaneous     80   270
```

0.3 Naive Analysis

If treatment had been randomized, the difference in the following point estimates would be a valid estimate of the causal effect of treatment:

```
naive_analysis <-
  dat %>%
  group_by(Surgery) %>%
  summarise(P_Success = sum(Success)/length(Success))
naive_analysis
```

```
## # A tibble: 2 x 2
##   Surgery      P_Success
##   <fct>            <dbl>
## 1 open             0.78
## 2 percutaneous     0.826
```

However, the observational nature of this data set has corrupted this comparison, resulting in the phenomenon known as "Simpson's Paradox". We see the "paradox" if we split things out by size of the kidney stone:

```
 dat %>% filter(Size == "small") %>%
  group_by(Surgery) %>%
  summarise(P_Success = sum(Success)/length(Success))
```

```
## # A tibble: 2 x 2
##   Surgery      P_Success
##   <fct>            <dbl>
## 1 open             0.931
## 2 percutaneous     0.867
```

```
 dat %>% filter(Size == "large") %>%
  group_by(Surgery) %>%
  summarise(P_Success = sum(Success)/length(Success))
```

```
## # A tibble: 2 x 2
##   Surgery      P_Success
##   <fct>            <dbl>
## 1 open             0.730
```

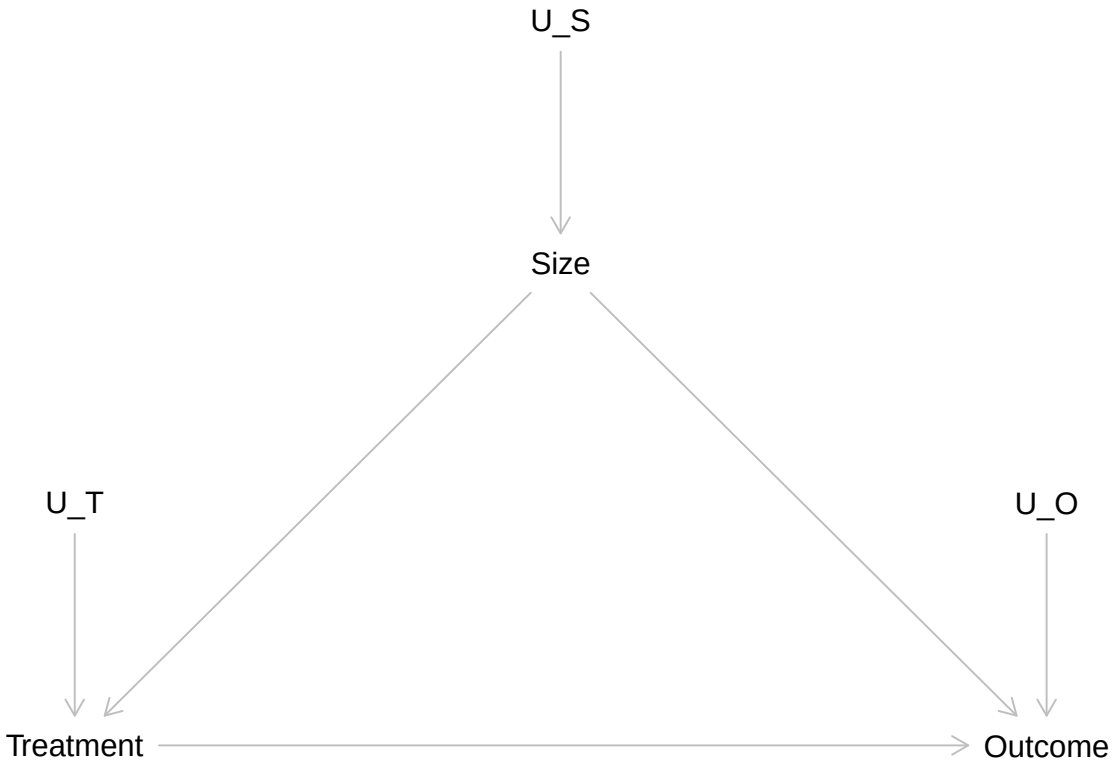
```
## 2 percutaneous     0.688
```

To summarize: based on point estimates,

- Open surgery has better efficacy for subjects with small stones,
- Open surgery has better efficacy for subjects with large stones,
- Each subject falls into one of those two categories . . . and yet:
- Point estimates from the naive analysis imply that percutaneous surgery is better "overall".

Intuitively, the naive "overall" analysis does not appear to be valid. As shown by Pearl et al., it is possible to formally prove the correctness of this intuition, given causal assumptions reflected by the following diagram :

```
scm <- dagitty('dag{
                  Treatment <- Size -> Outcome <- Treatment
                  U_T -> Treatment
                  U_S -> Size
                  U_O -> Outcome
                  }')
coordinates(scm) <- list(
 x = c(U_T = -1,  Treatment = -1, Size = 0, Outcome = 1, U_S = 0, U_O = 1 ),
 y = c(U_T = -sqrt(3)/2, Treatment = 0, Size = -sqrt(3),
       Outcome = 0, U_S = -sqrt(3)*1.5, U_O = -sqrt(3)/2)
)
plot(scm)
```



The "U" variables in the diagram represent possible unmeasured covariates. The essential point with respect to the unmeasured variables is that none of them exert causal influence on both treatment and outcome; the only confounding variable that *does* exert causal influence on both treatment and outcome is stone size, which is measured.

As noted in the primary manuscript, the arrows in causal diagrams are understood to convey truly causal (and not merely probabilistic) dependence. More precisely, the *lack* of an arrow from B to A is intended to convey that exogenous manipulation of B would not result in any changes in A. With the benefit of these more-than-probabilistic assumptions, the correct estimate of the causal effect can formally be shown to be equivalent to a weighted average of the conditional probabilities:

```
cond_probs <-
  bind_cols(
    dat %>% filter(Size == "large") %>%
      group_by(Surgery) %>%
      summarise(P_Success = sum(Success)/length(Success)) %>%
      select(-Surgery),
    dat %>% filter(Size == "small") %>%
      group_by(Surgery) %>%
      summarise(P_Success = sum(Success)/length(Success)) %>%
      select(-Surgery)
  ) %>%
  as.matrix()

cov_dist <-
  dat %>%
  group_by(Size) %>%
  summarise(P_Size = n()/nrow(dat)) %>%
  pull("P_Size")

simple_analysis <- cond_probs %*% cov_dist
simple_analysis <- data.frame(
  Surgery = levels(dat$Surgery),
  P_Success = simple_analysis
)
simple_analysis
```

```
##        Surgery P_Success
## 1         open 0.8325462
## 2 percutaneous 0.7788750
```

0.4 Pharmacometric Population Simulation

The same underlying logic is employed (and the same causal validity is therefore obtained) if we approach the problem via population simulation, following a routine "pharmacometrics" workflow:

1. Fit a model that takes into account the known explanatory factors affecting the response.
2. Create a virtual population with a covariate distribution reflecting the empirical covariate distribution.
3. Use the model to simulate the outcome for each subject in the virtual population if s/he receives open surgery. Compute the proportion of successes.
4. Use the model to simulate the outcome for each subject in the virtual population if s/he receives percutaneous surgery. Compute the proportion of successes.
5. Compare the proportions of success computed in the previous two steps.

In this application, the procedure may be represented in code as:

```r
model <- dat %>%
  group_by(Surgery, Size) %>%
  summarise(P_Success = sum(Success) / length(Success))

nsim <- 1e6

cov_dist <-
  dat %>%
  group_by(Size) %>%
  summarise(P_Size = n()/nrow(dat))

virtual_pop <-
  data.frame(
    SUBJID = 1:(nsim),
    Size = rep(cov_dist$Size,
               ## each covariate level represented the expected number
               ## of times in a population of nsim subjects:
               times = cov_dist$P_Size * nsim
               )
  )
```

```r
pop_sim_open <-
  virtual_pop %>%  mutate(Surgery = "open") %>%
  left_join(model) %>%
  ## If we were clever we could skip the following step of adding residual
  ## variability, but let's not use our brains as a crutch :-)
  mutate(SimSuccess = rbinom(n(), 1, P_Success)) %>%
  group_by(Surgery) %>%
  summarise(P_Success = sum(SimSuccess)/length(SimSuccess))
pop_sim_percutaneous <-
  virtual_pop %>% mutate(Surgery = "percutaneous") %>%
  left_join(model) %>%
  mutate(SimSuccess = rbinom(n(), 1, P_Success)) %>%
  group_by(Surgery) %>%
  summarise(P_Success = sum(SimSuccess)/length(SimSuccess))
pmx_analysis <- bind_rows(pop_sim_open, pop_sim_percutaneous)
pmx_analysis
```

```
## # A tibble: 2 x 2
##   Surgery      P_Success
##   <chr>            <dbl>
## 1 open             0.832
## 2 percutaneous     0.779
```

That gives us the same answer as the previous "simple" analysis, modulo some Monte-Carlo error from the simulation.

0.5 Comparison to recognized causal inference procedures.

The `simple` analysis and the `pmx` (pharmacometric population simulation) analysis above both rely on the same logic as the causal inference procedure known as *G-computation*. All three procedures assume the adequacy of the outcome model in the sense that the dependence of the

outcome on confounders is assumed to be well-characterized by the model. An ostensibly very different approach is to develop a model to predict the treatment that a subject will receive (or a "propensity model"), and make weighting adjustments based on that. The propensity-based approach will result in correct estimates of the causal effect as long as the dependence of treatment assignment on confounders is well characterized by the model (even if the dependence of outcome on confounders is not well characterized).

First note that we can think of our earlier "naive" analysis as the application of a weighted procedure where subject received a weight of 1:

```
dat %>%
  mutate(Weight = 1) %>%
  group_by(Surgery) %>%
  summarise(P_Success = sum(Success * Weight)/sum(Weight))
```

```
## # A tibble: 2 x 2
##   Surgery      P_Success
##   <fct>            <dbl>
## 1 open             0.78
## 2 percutaneous     0.826
```

The inverse probability weighting (IPW) approach instead uses weights related to propensity scores:

```
prop_scores <-
  dat %>%
  group_by(Size) %>%
  summarise(Prop_Score = sum(Surgery == "percutaneous") / length(Surgery))

ipw_analysis <-
  dat %>%
  left_join(prop_scores) %>%
  mutate(Weight = case_when(
    .$Surgery == "open" ~ (1 / (1-Prop_Score)),
    .$Surgery == "percutaneous" ~ (1 / Prop_Score)
  )) %>%
  group_by(Surgery) %>%
  summarise(P_Success = sum(Success * Weight)/sum(Weight))
```

```
compare <- bind_cols(
  naive = naive_analysis %>% rename(Naive = P_Success),
  simple = simple_analysis %>% select(-Surgery, Simple = P_Success),
  pmx = pmx_analysis %>% select(-Surgery, PMX = P_Success) ,
  ipw = ipw_analysis %>% select(-Surgery, IPTW = P_Success)
)
compare
```

```
## # A tibble: 2 x 5
##   Surgery      Naive Simple   PMX  IPTW
##   <fct>        <dbl>  <dbl> <dbl> <dbl>
## 1 open         0.78   0.833 0.832 0.833
## 2 percutaneous 0.826  0.779 0.779 0.779
```

0.6 Conclusions

The foregoing comparison is intended to illustrate that pharmacometric population simulation (represented in the code as the `pmx` analysis) is a valid causal inference procedure under the same assumptions that would justify *G-computation* as a valid causal inference procedure. Under those assumptions, pharmacometric population simulation also results in estimates equal in value to those obtained by inverse propensity weighting. In other words, estimates based on pharmacometric population simulation will not suffer from the confounding associated with "Simpson's Paradox" as long an adequate outcome model is employed.